

Hasiniaina Christian LOVANIRINA

Full-Stack Engineer | Applied AI, RAG & Automation

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GitHub: <https://github.com/hasiniaina7> | LinkedIn: <https://www.linkedin.com/in/hasiniaina-christian/> | Portfolio: <https://hasiniaina7.github.io/>

Availability: international remote collaboration through a compliant arrangement suited to the role.

Profile

Confirmed Full-Stack Developer specializing in operational applied AI. Designs traceable RAG, agent, webhook and automation systems with confidence thresholds, fallback, human handoff, security, tests and controlled production rollout.

Core skills

Applied AI and automation: AI agents, RAG, OpenAI API, n8n, OpenClaw, Meta Webhooks, ChromaDB, embeddings, AI traces, confidence scoring, fallback and human handoff.

Backend and APIs: Python, Django, Django Ninja, Node.js, Next.js API, Express.js, PostgreSQL, Supabase, Prisma, Sequelize, REST, OpenAPI, webhooks, workers and outbox.

Data, security and quality: PostgreSQL, RLS, RBAC, multi-tenancy, audit logs, HMAC, AES-256-GCM, secrets, pytest, Vitest, Playwright, Docker, Linux and runbooks.

Web product: React, TypeScript, Next.js, Vite, PWA, dashboards, B2B back offices, technical SEO, public tracking, complex forms and local synchronization.

Professional experience

TechZone IT Solution | Confirmed Full-Stack Developer / Technical Lead

October 2024 - present | Fianarantsoa, Madagascar

B2B social-commerce orchestrator | <https://www.iagasy.com/>

- Built a multi-tenant Next.js/Supabase/PostgreSQL SaaS with Meta webhooks, message queues, workers, an agent endpoint, a knowledge base and n8n automation.
- Delivered a RAG pipeline with ingestion, chunking, embeddings, ChromaDB, intent detection, reranking, confidence thresholds, fallback and human handoff.
- Secured flows with X-Hub-Signature-256, AES-256-GCM token encryption, deduplication, stale-event rejection and controlled go-live scripts.

TZ Smart - network operations, field service and payment SaaS | <https://tzsmart.app>

- Designed a modular Django Ninja/PostgreSQL backend by business domain, with contract-first APIs, OpenAPI, RBAC, validation and regression tests for web and Android clients.
- Delivered a React/Vite back office and Kotlin Android application covering network diagnostics, field operations, offline/replay, FCM and MVola payments.
- Operated production with health checks, watchdogs, runbooks, sensitive-data protection and recovery procedures.

SMS Payment Gateway - secured MVola Android gateway

- Designed device enrollment with an Android Keystore-sealed secret, HMAC-signed requests and protected gateway endpoints.
- Implemented MVola SMS parsing, sender filtering, deduplication, Room/SQLCipher, inbound/outbound queues, heartbeat and recovery through Foreground Service, WorkManager and BootReceiver.

Fretunia / Trackmada - multi-tenant B2B logistics SaaS | <https://fretunia.com>

- Built a multi-tenant Supabase/PostgreSQL architecture with RLS, RBAC, memberships, API keys, webhooks, audit logs and tenant isolation.
- Delivered parcel, batch, PDF invoicing, pricing, public tracking, customer sites, custom domains and outbox/n8n notification modules.
- Contributed to a platform used by more than 20 companies in Africa, with auth, billing, tenant, RLS and cross-tenant tests.

Malagasy speech-synthesis R&D - XTTS v2

- Explored feasibility, prepared a voice dataset, ran XTTS v2 fine-tuning trials and documented data, audio-quality and GPU-cost limitations.

AIM Madagascar | Full-Stack Web/Mobile Developer | <https://dms-aim.online/>

October 2023 - present | Madagascar | Development contract

- Designed Express/Sequelize/PostgreSQL and REST API flows for beneficiaries, cashiers, payment periods, payments, adjustments, post-closing corrections and business controls.
- Implemented controlled financial exports, sensitive-action audit and safeguards for workflows handling confirmed volumes from 500M+ MGA to 2B+ MGA as system criticality.

Infrastructure and field foundation

- **Action Against Hunger International | IT Infrastructure Officer | January 2024 - June 2024:** networks, Microsoft 365, backups, cybersecurity, L2/L3 support and field operations.

Professional method

Structured use of development agents to accelerate implementation, testing and documentation, with specifications, human review, security validation and final accountability.

Education and languages

Bachelor-level degree in Computer Science (Bac+3) | 2016 - 2022

French: fluent | **English:** B2, professional proficiency